

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____ 129

Does R2 show up as an OSPF neighbor on R1? _Yes_

Does R2 show up as an OSPF neighbor on R3? _No_

What does this information tell you?

This information tells us that all traffic from R3 to the 192.168.2.0/24 network is now being routed through R1. This happens because the S0/0/1 interface on R2 is still configured as a passive interface, so OSPF routing updates are not sent across this link. The accumulated cost of 129 is calculated because the traffic must now pass through two serial links (cost of 64 each) plus the GigabitEthernet LAN link on R2 (cost of 1).

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The accumulated cost metric is 65.

It is calculated by adding the cost of the direct serial link between R3 and R2 (which has a default cost of 64 for a T1 link) and the cost of the GigabitEthernet 0/0 LAN interface on R2 (which has a cost of 1). $64 + 1 = 65$.

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To obtain a more accurate cost calculation for these faster links (higher than 100 Mb/s).

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: R1 S0/0/1 + R3 G0/0 ($781+1=782$).

192.168.23.0/30: R1 S0/0/1 + R3 S0/0/1 ($781+64=845$)

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562.

Each serial link has a cost of 781, and the route to the 192.168.23.0/24 (maybe /30?) network travels over two links, therefore it is 1562 in total.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because OSPF will select the route with the least cost. The cost of the route through R2 is less than the route direct from R1 to R3

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Because Designated Router and Backup Designated Router election needs Router ID assignments. If the router ID is default (using the physical interface IPO), the ID could change if that interface goes down, and this would cause time-consuming OSPF recalculations and instability. It also makes troubleshooting easier.

2. Why is the DR/BDR election process not a concern in this lab?

Because the routers in this lab are connected to each other using serial interfaces, which are point-to-point links. The OSPF DR/BDR election process only occurs on multi-access networks (like Ethernet broadcast networks), not on point-to-point networks.

3. Why would you want to set an OSPF interface to passive?

Because it prevents OSPF Hello packets from being sent out of that interface. This can save bandwidth and other computational resources, such as CPU on LAN interfaces, where no other routers exist. It also improves security by preventing unauthorized devices.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

Area.

Dividing an OSPF domain into multiple areas helps localize routing updates through hierarchical design, reduces the size of routing tables, and minimize the impact of topological changes.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be used. When a router learns about the exact same destination network from different routing protocols, it uses Administrative Distance (AD) to choose the best route, not the metric. EIGRP has the lowest AD (90), compared to OSPF (110) and RIP (120).